INCH-POUND

MIL-DTL-81905E(AS)
21 December 2005
SUPERSEDING
MIL-A-81905D(2)(AS)
30 October 1999

DETAIL SPECIFICATION

ANTI-G GARMENT, CSU-15A/P

This specification is approved for use by the Naval Air Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements and inspections for the Anti-G Garment, CSU-15A/P.

1.2 Classification. The Anti-G Garment, CSU-15A/P, furnished under this specification will be one of the following sizes (see 6.2).

Small regular	Medium regular	Large regular
Small long	Medium long	Large long
		Large extra long

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents citied in other sections of this specification or recommended for additional information or examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents citied in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to the Commander, Naval Air Warfare Center Aircraft Division, Code 491000B120-3, Highway 547, Lakehurst, NJ 08733-5100 or emailed to thomas.omara@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

V-T-295	-	Thread, Nylon

FEDERAL STANDARDS

FED-STD-4	-	Glossary of Fabric Imperfections
FED-STD-595	-	Colors Used in Government Procurement

COMMERCIAL ITEM DESCRIPTIONS (CIDs)

A-A-3174	-	Plastic Sheet, Polyolefin
A-A-50195	-	Thread, Aramid
A-A-55126	-	Fastener Tapes, Hook and Loop, Synthetic
A-A-55621	-	Fasteners, Snap, Style 3 Pronged Ring Head Type (16 LIGNE)
A-A-55634	-	Zippers (Fasteners, Slide Interlocking)
A-A-55809	-	Insulation Tape, Electrical, 600V, Polyvinyl Chloride, Pressure-Sensitive Adhesive

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-R-6855	-	Rubber, Synthetic, Sheets, Strips, Molded or Extruded Shapes, General Specification For
MIL-DTL-10884	-	Fasteners, Snap
MS27755	-	Connector, Hose Quick Disconnect, Female, Anti-G Suit
MIL-DTL-32075	-	Label: for Clothing, Equipage, and Tentage, (General Use)
MIL-T-38328	-	Tape, Textile, Nylon, Aromatic, Non-melting, Reinforcing
MIL-C-43204	-	Cloth, Spacer (Olefin)
MIL-L-43283	-	Leather, Calfskin, Kip, and Cattlehide for Footwear Uppers, Chrome Tanned
MIL-W-81116	-	Webbing, Textile, Polyamide, High Temperature Resistant, Loop
MIL-C-83242	-	Cord, Aromatic Polyamide, Non-melting (Inactive for new design)
MIL-C-83390	-	Connector, Hose, Quick-Acting, Male, Anti-G Garment (Inactive for new design)

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MIL-C-83429	-	Cloth, Plain and Basket Weave, Aramid
MIL-C-83489	-	Cloth, Coated, Nylon, Polyurethane Coated

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL AIR SYSTEMS COMMAND

1628AS100	-	Top Assembly Drawing, Anti-G Garment CSU-15A/P
1628AS101	-	Patterns, CSU-15A/P Anti-G Garment, Size Small Regular
1628AS102	-	Patterns, CSU-15A/P Anti-G Garment, Size Small Long
1628AS103	-	Patterns, CSU-15A/P Anti-G Garment, Size Medium Regular
1628AS104	-	Patterns, CSU-15A/P Anti-G Garment, Size Medium Long
1628AS105	-	Patterns, CSU-15A/P Anti-G Garment, Size Large Regular
1628AS106	-	Patterns, CSU-15A/P Anti-G Garment, Size Large Long
1628AS107	-	Patterns, Anti-G Garment, CSU-15A/P, Size Large Extra Long
1628AS108	-	Patterns, CSU-15A/P Anti-G Garment, Miscellaneous All Sizes
1628AS900	-	Spacer, Tube – Anti-G Garment
3926AS100	-	Fasteners, Slide, Anti-G Garment

(Copies of these drawings are available from Defense Supply Center Philadelphia, Clothing and Textiles Directorate, Attn: DSCP-CBTC, (Bldg 6), 700 Robbins Ave., Philadelphia, Pa. 19111-5092 or through the online technical data request system, http://warfighter.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS397	-	Clamp - Ratchet, One Piece. (DoD adopted)

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NASM20230	-	Grommets, Metallic, Plain and Spur, With Washers, Type I and Type III. (DoD adopted)
NASM27980	-	Fasteners, Snap, Style 2 (Regular Wire Spring Clamp Type). (DoD adopted)
NASM27983	-	Fasteners, Snap, Style 4 (Three Way Locking Type). (DoD adopted)

(Copies of these documents are available online at http://www.aia-aerospace.org or from the Aerospace Industries Association, 1000 Wilson Blvd, Suite 1700, Arlington, VA 22209-3901.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQC-Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes. (DoD adopted)

(Copies of this document are available online at http://www.asq.org or from the American Society for Quality, P.O. Box 3005, 611 E. Wisconsin Avenue, Milwaukee, WI 53201-4606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) INTERNATIONAL

ASTM-A313/A313M	-	Wire, Stainless Steel Spring. (DoD adopted)
ASTM-D1974	-	Fiberboard Boxes, Methods of Closing, Sealing and Reinforcing. (DoD adopted)
ASTM-D2261	-	Fabrics, Woven, By the Tongue (Single Rip) Method (Constant-Rate-of-Extension Tensile Testing Machine), Tearing Strength of. (DoD adopted)
ASTM-D3775	-	Fabric, Wrap End Count and Filling Pick Count of Woven. (DoD adopted)
ASTM-D3776	-	Fabric, Mass Per Unit Area (Weight) of. (DoD adopted)
ASTM-D5034	-	Fabrics, Textile (Grab Test), Breaking Strength and Elongation of. (DoD adopted)
ASTM-D6193	-	Stitches and Seams. (DoD adopted)

(Copies of these documents are available online at http://www.astm.org or from ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

THE PARACHUTE INDUSTRY ASSOCIATION (PIA)

PIA-C-3953	-	Cloth, Duck, Nylon. (DoD adopted)
PIA-W-4088	-	Webbing, Textile, Woven, Nylon. (DoD adopted)

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(Copies of these documents are available online at http://www.pia.com/ or from the Parachute Industry Association, 3833 West Oakton St., Skokie, IL 60076.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE) WORLD HEADQUARTERS

SAE-AMS-T-22085 - Tapes, Pressure Sensitive, Adhesive, Preservation and Sealing. (DoD adopted)

(Copies of this document are available online at http://www.sae.org or from SAE World Headquarters, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Anti-G Garments furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.3 and 6.6).

3.2 First article. When specified (see 6.2) samples shall be subjected to first article inspection in accordance with 4.4.

3.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials shall be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Materials and components. Except for the metallic materials and parts, unless otherwise specified the materials and components shall have been manufactured not greater than twelve months prior to the date of delivery of the Anti-G Garment.

3.4.1 Basic materials.

3.4.1.1 High temperature resistant cloth. Except for the bladder, the basic fabric for the Anti-G Garment shall conform to MIL-C-83429, Type II and shall be in accordance with FED-STD-595, sage green 1590.

3.4.1.2 Polyurethane coated nylon cloth. The material for the bladder (including the attachment patches) and reinforcement of the inflation external tube covering shall conform to MIL-C-83489, Type I.

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3.4.2 Reinforcements and bindings. Unless otherwise specified herein, the reinforcements shall be made of the basic fabric specified in 3.4.1.1. The 5/16-inch-wide (finished), centerfold, bias binding shall be 45 degrees bias cut 1-1/4 inches to 1-3/8 inches wide from the basic fabric specified in 3.4.1.1.

3.4.3 Spacer material. The spacer material for the bladder and bladder inflation tube shall conform to MIL-C-43204, Type III.

3.4.4 Adjustment laces and lanyard. The cord for the lanyard and for the adjustment laces shall conform to MIL-C-83242, Type I and shall approximately match the shade of the basic material in 3.4.1.1.

3.4.5 Slide fasteners. The slide fasteners (zippers) shall be brass with a short tab pull. The slide fasteners shall conform to tables I and II of this specification and to A-A-55634 and drawing 3926AS100 except as otherwise specified herein. The slide fasteners for the body opening shall be Type II , Style 7A, Size MHS and for the leg openings shall be Type III, Style 7A & 8A, Size MHS with S-25 retainer and pin assembly, metal 0308 with a black oxidized finish. The slide fasteners for the thigh take-ups shall be Type I, Style 4, Size MS and for the pockets shall be Type I, Style 7, Size MS. A substitute slide fastener for the body opening, the leg openings, and the thigh take-ups shall be used only when specifically approved by the contracting officer. The tape and the bead of all slide fasteners (including the body opening) shall be a high temperature resistant polyamide material that has been dyed to match the color of the basic fabric specified in 3.4.1.1. The tape of the slide fastener shall have color fastness in accordance with A-A-55634 except that “fair” colorfastness in lieu of “good” color fastness to light will be acceptable. Slide fasteners from a supplier not identified on the slide fastener Engineering Support Activity (ESA) source control drawing 3926AS100 shall not be used without prior approval of the ESA. The contracting officer shall be contacted for any substitution requests. The finish of the chain and the other metal components shall be black. All automatic-locking sliders shall have a pin-type locking device. A pull tape (see 3.4.7) shall be permanently fastened to each short tab pull with two bartacks.

TABLE I. Slide fasteners.

Location of Slide Fastener	Quantity	CID A-A-55634		
		Type	Style	Size
Body opening	1	III	7A	MHS
Leg opening	1/	III	7A and 8A	MHS
Thigh take-ups	2	I	4	MS
Pockets	2	I	7	MS

1/ One of each style.

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3.4.6 Stiffeners. The stiffeners for the slide fasteners and the stiffeners for the front body and the back body shall be any subdued color made of nylon cloth that conforms to PIA-C-3953, Class 2.

3.4.7 Aromatic polyamide webbing. The webbing for the pull tape for the slide fasteners, the loops for the slide fasteners, and the hanger shall be 5/16 inch wide, shall conform to MIL-T-38328, Type VI, and shall approximately match the color of the basic fabric specified in 3.4.1.1.

3.4.8 Nylon webbing. The webbing for the knife pocket reinforcement shall be untreated, shall conform to PIA-W-4088, Type XII, and shall approximately match the shade of the basic material in 3.4.1.1 or be natural. The webbing for the stiffener shall conform to PIA-W-4088, Type I.

TABLE II. Length of slide fasteners.

Size of Anti-G Garment	Length (Inches of Slide Fasteners)			
	Body Opening 1/	Leg Opening 2/	Thigh Take-ups	Pockets
Small Regular	8-1/2	26	10-1/2	8
Small Long	8-1/2	28	11-1/2	8
Medium Regular	8-1/2	26-1/2	11	8
Medium Long	8-1/2	28-1/2	11-1/2	8
Large Regular	8-1/2	27	11	8
Large Long	8-1/2	29	11-1/2	8
Large Extra Long	8-1/2	33	11-1/2	8

1/ The tolerance for the length of the slide fastener for the body opening shall be ± 1/8 inch.

2/ The tolerance for the length of the slide fastener for the leg opening shall be ± 1/4 inch.

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3.4.9 Wrapping for Anti-G hose assembly. The tape for wrapping the inflation tube assembly shall be black, 1 inch wide tape conforming to A-A-55809.

3.4.10 Lacer loops. The tape for the lacer loops shall conform to MIL-W-81116 except that the sleeve may have 16 carriers in lieu of 20 carriers, the length shall be 39-1/2 inches in lieu of 45 inches for each yard of webbing, and the color shall match the color of the basic fabric as specified in 3.4.1.1.

3.4.11 Thread. Unless otherwise specified herein, the thread for all sewing operations shall conform to Size E, Soft or Bonded, 600 Denier, Tex 60, 3 Ply, 6 turns per inch, 5 lbs minimum break strength, 38 percent elongation conforming to A-A-50195 and shall approximately match the shade of the basic material in 3.4.1.1. (Note: This thread may be used to sew the nylon mesh cloth (see 3.4.21) in lieu of the thread conforming to V-T-295, Type I, Class 1, Size E which is specified on the drawing.)

3.4.12 Snap fasteners. Except for the knife pocket, the snap fasteners shall conform to A-A-55621, Style 3, Finish 2 (pronged ring A-A-55621-1B, socket A-A-55621-2B, stud A-A-55621-4B, and eyelet A-A-55621–5B). The snap fastener for the knife pocket shall conform to NASM27983, Style 4, Construction A (socket NASM27983-2, button NASM27983-1, stud NASM27983-3, and eyelet NASM27983-4).

3.4.13 Grommet. The grommet shall conform to NASM20230, Size 00 and shall be oxidized a black color.

3.4.14 Fastener tape. The hook fastener tape shall conform to A-A-55126, Type I, Class 2, except for the color. The pile fastener tape shall conform to A-A-55126, Class 2, except for the color. The color of the pile fastener tape and the hook fastener tape shall approximately match the color of the basic fabric specified in 3.4.1.1. Except for the pile fastener tape for the hose retainer and the checklist retainer, the fastener tape shall be 1 inch wide. The pile fastener tape for the hose retainer and the checklist retainer shall be 2 inches wide.

3.4.15 Adhesive. The adhesive for cementing the spacer material to the coated bladder material shall be a UR-1092, Clifton Adhesive Part Number 7531037, 282 Burgess Place, Wayne, NJ 07474-0282.

3.4.16 Connector. The connector for the inflation tube shall conform to MIL-C-83390.

3.4.17 Rubber sleeve. The rubber sleeve for the connector shall be tubing conforming to MIL-R-6855, Class 2, Grade 40 or 60, shall be 3 $\pm$ 1/16 inches long; shall have an outer diameter of 3/4 $\pm$ 1/16 inches and shall have a wall thickness of 1/8 $\pm$ 1/32 inches. No silicone or any other lubricant shall be applied to either the rubber sleeve or the connector prior to assembling the rubber sleeve to the connector.

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3.4.18 Clamps. The clamps for securing the connector to the inflation tube and external inflation tube covering shall conform to NAS397-14 and NAS397-16.

3.4.19 Leather reinforcement. The leather for the reinforcement patch over the inflation sleeve opening and knife pocket reinforcement shall be black, and shall conform to MIL-L-43283, Type I, except that the thickness (ounces) shall be 2 to 2-1/2 ounces.

3.4.20 Springs for spacer tube. The material for the springs of the spacer tube shall be 0.016 inch diameter wire that conforms to ASTM-A313/A313M, Form II, Composition 304, Condition B.

3.4.21 Cloth for the spacer tube. The cloth for the spacer tube shall be polypropylene netting/mesh cloth with approximately 14x12 strand count per inch, hole size 0.05 x 0.08 inches, 0.020 thick, and 19 pounds per 1000 feet squared.

3.4.22 Rubber band. The rubber band (see figure 1) shall be 3/4 $\pm$ 1/16 inches long, and 0.035-inch minimum thickness. The color of the rubber band shall be black. The rubber shall comply with MIL-R-6855, Class 2, Type A.

3.5 Design. The Anti-G Garment shall be a wrap-around garment that extends from the waist to over the calves of the wearer. It shall have an inflatable bladder that will cover the waist and the leg areas of the wearer and a connector that will provide normal and emergency connection and disconnection of the bladder of the Anti-G Garment to and from the air pressure source. Both sides of the Anti-G Garment shall have external adjustment laces in the waist, thigh, and calf areas; however, the adjustment laces shall have covers that can be unfastened for easy accessibility. The Anti-G Garment shall have a pocket on the calf of each leg and a knife pocket on the front of the left thigh.

3.6 Construction. The Anti-G Garment shall be constructed as specified in table III; however, the manufacturer shall not be required to follow the exact sequence of operations as listed therein.

3.6.1 Stitches, seams, and stitching. Stitches, seams, and stitching specified in table III shall conform to ASTM-D6193. The looper (under) thread of stitch Type 401 shall be on the inside of the Anti-G Garment. Thread breaks and ends of all seams and stitching, if not caught in other seams or stitching, shall be securely backstitched by reversing the stitching direction and sewing a continuous row of stitching superimposed on the original row of stitching for a distance of 1/2 inch. A stitching shall be considered caught in another stitching or seam if the two stitches are in opposite directions and superimposed directly on each other for at least 1 inch. The bartacks joining the spacer material and attaching the patches as specified in figures 2 and 3 shall be 3/4 inch long, centered, and shall contain approximately 32 stitches. All thread ends inside and outside of the Anti-G Garment shall be trimmed flush with the fabric.

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TABLE III. Sewing operations.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
1	Cut Parts (see 3.11.1) (a) Cut the parts (including the bladders) of the Anti-G Garment in accordance with the patterns which show shape, size, directional lines, and marking for the proper assembly of the parts. Note: Trimming of the body lacer covers, the thigh lacer covers, and the calf lacer covers will be permitted as specified in No. 5(b), 6(b), and 6(e).			
2	Shade Mark (a) Mark all parts to ensure a uniform shade throughout the Anti-G Garment.			
3	Sear Raw Edges (a) Sear the raw edges of all parts made of nylon fiber.			
4	Sew Loops of Lacer Loop Tape (a) Sew one row of stitching along each length of the lacer loop tape, through the base of each loop to prevent the loops from slipping.	301	SSa-1	10-12

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
5	Make Body Lacer Covers Note: No. 5(b) shall be performed before No. 5(c) so that the fastener tape will be sewn to the body lacer covers before the body lacer covers are bound. (a) Select the fastener tapes for the body lacer covers in accordance with table IV. (b) Trim the fastener tape, if necessary, to fit the body lacer covers at the top and the bottom. Turn each body lacer cover under 3/8-inch. Sew the hook fastener tape to each body lacer cover, at the pattern location, with a single row of stitching 1/16 inch from the edge on all four sides.	301	LSd-1	10-12
	(c) Bind each body lacer cover at each end with 5/16-inch-wide (finished) bias binding 1/16 ± 1/32 inch from the edge. Before completing the stitching of the binding at the top and the bottom of each lacer cover, fold the raw edges of each end of the binding (two ends per lacer cover) approx. 1/2 inch, so no raw edges of binding tape are exposed, and complete the stitching of the binding to the fastener tape.	301	BSc-1	10-12

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
6	Make Leg (thigh and calf) Lacer Covers Note: No. 6(b) shall be performed before No. 6(c) so that the fastener tape will be sewn to the thigh lacer covers before the thigh lacer covers are bound. Note: No. 6(e) shall be performed before No. 6(f) so that the fastener tape will be sewn to the calf lacer covers before the calf lacer covers are bound. (a) Select the fastener tapes for the thigh lacer covers in accordance with table IV. (b) Trim the fastener tape, if necessary, to fit the thigh lacer covers at the top and the bottom. Turn each thigh lacer cover under 3/8 inch. Sew the hook fastener tape to each thigh lacer cover, at the pattern location, with a single row of stitching 1/16 inch from the edge on all four sides.	301	LSd-1	10-12
	(c) Bind each thigh lacer cover at each end with 5/16-inch-wide (finished) bias binding 1/16 ± 1/32 inch from the edge. Before completing the stitching of the binding tape at the top and the bottom of each lacer cover, fold the raw edges of each end of the binding (two ends per lacer	301	BSc-1	10-12

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
6	cover) approximately 1/2 inch, so that no raw edges are exposed, and complete the stitching of the binding to the fastener tape. (d) Select the fastener tapes for the calf lacer covers in accordance with table IV. (e) Trim the fastener tape, if necessary, to fit the calf lacer covers at the top and the bottom. Turn each calf lacer cover under 3/8 inch. Sew the hook fastener tape to each calf lacer cover, at the pattern location, with a single row of stitching 1/16 inch from the edge on all four sides.	301	LSd-1	10-12
	(f) Bind each calf lacer cover at each end with 5/16-inch-wide (finished) bias binding 1/16 ± 1/32 inch from the edge. Before completing the stitching of the binding at the top and the bottom of each lacer cover, fold the raw edges of each end of the binding (two ends per lacer cover) approximately 1/2 inch so that no raw edges are exposed, and complete the stitching of the binding to the fastener tape.	301	BSc-1	10-12
7	Hem Body Bladder Cover (a) Hem the upper edge of the separate leg extension to the inside, 1/4 inch, with a single	301	EFa-1	10-12

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
7	row of stitching 1/16 inch from the edge. (b) Fold the end of the outside body bladder cover leg, to which the separate leg extension is to be attached to the inside 1/4 inch, and hem with a single row of stitching 1/16 inch from the edge. Before completing the stitching of the binding at the top and the bottom of each lacer cover, fold the raw edges of each end of the binding (two ends per lacer cover) approximately 1/2 inch so that no raw edges are exposed, and complete the stitching of the binding to the fastener tape. Note: The terms "body bladder cover", "front body bladder case", "outside cover", and "forward cover" as indicated in this specification and on the pattern drawings of the bladder cover shall be interpreted as that portion of the bladder cover on the outside (away from the wearer's body) and has a truncated section cut off. The terms "back body bladder cover "or "case" shall be interpreted as the cover positioned next to the wearer's body and is full size as indicated on the pattern drawings. (c) Lap the hemmed end of the separate leg extension up under the body bladder cover leg	301 301	EFa-1 SSa-1	10-12 10-12

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
7	end until the lower edge is even with the opposite leg end, and sew at the sides with a single row of stitching 1/8 inch from the edge.			
8	Join Bladder Case			
	(a) Join the front and the back body bladder case, material face to face, with a single row of stitching 1/4 inch from the edge, leaving the legs, the bladder inflation sleeve, and 1 inch at tab locations (top, bottom, and right side) open.	301	SSe-2(a)	10-12
	(b) Turn and topstitch with a single row of stitching 1/16 inch from the edge.	301	SSe-2(b)	10-12
9	Hem Leg Bladder Covers			
	(a) Hem the leg bladder covers to the inside, 1/4 inch, down the sides and around the bottom and the top, with a single row of stitching 1/16 inch from the edge.	301	EFa-1	10-12
10	Make Thigh Take-Ups			
	(a) Select the slide fasteners for the thigh take-ups in accordance with table II. Zigzag stitch the tapes of the slide fasteners together from the bottom stop to the ends of the tapes.	301 or 401	SSa-1	20-26

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
10	(b) Sew the 3/4 inch wide stiffener to the inside of the thigh take-up fly, right and left, on the long center line from end to end with a single row of stitching.	301	SSa-1	10-12
	(c) With the open end of the slide fastener at the top, center the top and the bottom stops between the top and the bottom of the thigh take-ups. Match the edges of the slide fastener tape with the edges of the thigh take-up's fly. Cut the ends of the slide fastener tapes off even with the cover, and sew all around with a single row of stitching 1/8 ± 1/16 inch from the edge.	301	SSa-1	4-6
	(d) Bind all around the thigh take-ups, catching the ends of the slide fastener tapes in the binding, with 5/16-inch-wide (finished) bias binding 1/16 ± 1/32 inch from the edge.	301	BSc-1	10-12
	(e) Thread a 5-inch length of the 5/16-inch-wide webbing for the pull tape through the pull of each slide fastener. Fold the webbing in the center, and bartack at the ends and close to the pull.	Bartack		21-23 stitches per bartack
11	Prepare Slide Fasteners for Body and Leg Openings			

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
11	(a) Select the slide fasteners for the body and the leg openings in accordance with table II.			
	b) Prefold the slide fastener tapes for the leg openings at the open ends with a triangular fold so that the beaded edge of the tape is across the end and the fold is to the inside. Sew with a single row of stitching 3/32 $^{+\ 1/32,}_{-\ 1/16}$ inch from the edge. Prefold and sew the slide fastener tape for the body opening in the same manner as for the leg openings, making certain that the fold is placed so that the length of the slide fastener tape will be 8-1/2 inches $\pm$ 1/8 inch).	301	EFa-1	10-12
	(c) Thread a 5-inch length of the 5/16-inch-wide webbing for the pull tape through the pull of each slide fastener. Fold the webbing in the center, and bartack at the ends and close to the pulls.	Bartack		21-23 stitches per bartack
	(d) Cut three 6-inch-long tapes from the 5/16-inch-wide webbing. Fold the tapes in the center to form a loop; join, forming a box and "X" with a single row of stitching 1/16 inch from the edge, to the separating end of the slide fastener opposite the sliders, with the ends of the tape down 5/8 inch from the end of the slide fastener	301	SSa-1	10-12

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
11	tape. Sew the tape to the inside of the Anti-G Garment.			
	(e) Sew the loops of the two tapes on the leg opening slide fasteners closed by sewing both sides of the loops with a single row of stitching 1/16 inch from the edges.	301		10-12
12	Make Body Protective Cover			
	(a) Place a 1-inch-square reinforcement of the basic fabric as marked on the pattern so that the center of the reinforcement will be 3/4 inch down from the top of the finished fly and on the inside, and sew with a single row of stitching 1/8 inch from the edge. Place another 1-inch-square reinforcement of the basic fabric so that the center of the reinforcement will be 5-1/2 inches below the center of the upper reinforcement and equally distant from the edge of the protective fly cover.	301	SSa-1	10-12
	(b) Fold the body protective fly cover lengthwise on the fold line, the material face to face; sew across the bottom and the top with a single row of stitching 3/8 inch from the edge.	301	SSe-2(a)	10-12

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
12	(c) Turn and insert the stiffener, folding the raw edge to the inside enclosing the stiffener. Sew all around the top, the bottom, and the two sides with a single row of stitching, 1/8 inch from the edge. Sew four additional rows of stitching, evenly spaced end to end.	301	SSe-2(b)	10-12
	(d) Sew two boxes "X" 7/8 x 7/8-inch, 1/8 inch from the edge of two reinforcements described in 12(a) with single row stitching.	301	SSa-1	10-12
	(e) Punch a hole diameter 1/8 inch in the center of each box "X" described in operation (d), install stud NASM27980-7B and eyelet NASM27980-8B through the holes at the center of box "X" stitch, ensuring the stud is on the side matching with the socket of tab assembly.			
	(f) Attach the slide fastener tape (with the slider attached) to the back edge of the body protective fly, the open end at the top; sew with a single row of stitching 1/8 inch from the edge. The slide fastener will extend below the bottom edge of the body protective fly by 5/8 $\pm$ 1/8 inch.			

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
13	Make Leg Protective Cover			
	(a) Position a 1-inch-square reinforcement of the basic fabric as marked on the patterns so that the center of the reinforcement will be 2-5/8 inches down from the top of the leg protective fly finished edge and on the inside. Sew with a single row of stitching 1/8 inch from the edge.	301	SSa-1	10-12
	(b) Fold the leg protective fly cover lengthwise on the fold line, the face sides of the material together; sew across the bottom and the top with a single row of stitching 3/8 inch from the edge.	301	SSe-2(a)	10-12
	(c) Turn and insert the stiffener, folding the raw edges to the inside enclosing the stiffener; sew all around the top, the bottom, and the two sides with a single row of stitching 1/8 inch from the edge. Sew four additional single rows of stitching, evenly spaced from end to end.	301	SSe-2(b)	10-12
	(d) Sew two boxes "X" 7/8 x 7/8-inch, 1/8 inch from the edge of two reinforcements described in 12(a) with single row stitching.	301	SSa-1	10-12
	(e) Punch a hole diameter 1/8 inch in the center of each box "X" described in operation (d), install			

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
13	stud NASM27980-7B and eyelet NASM27980-8B through the holes at the center of box "X" stitch, ensuring the stud is on the side matching with the socket of tab assembly.			
	(f) Attach the slide fastener tape (with the slide attached) to the back edge of the leg protective fly, the open end at the bottom; sew with a single row of stitching 1/8 inch from the edge.	301	SSa-1	10-12
14	Assemble Pockets			
	(a) Bind the side edges of the pockets with 5/16-inch-wide (finished) bias tape with a single row of stitching 1/16 ± 1/32 inch from the edge.	301	BSc-1	10-12
	(b) Place the slide fastener at the marked location on the pocket to the inside; join with a single row of stitching 1/16 inch from the edge all around. (The bridge end shall be at the lowest point on the pocket opening.)	301	SSa-1	10-12
	(c) Slit through the center and diagonally at the ends. Turn the fabric under approximately 1/4 inch; sew with a single row of stitching 1/16 inch from the folded edge.	301	LSd-1	10-12

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
14	(d) Sew the bottom corners, the face sides of the material together, with a single row of stitching 3/8 inch from the edge.	301	SSa-1	10-12
	(e) Topstich the sides and the bottom edge of the pockets, at the pattern marks, with a single row of stitching 1/16 inch from the folded edge through leather reinforcement patch.	301	OSf-1	10-12
	(f) Thread a 5-inch length of the 5/16-inch-wide-webbing for the pull tape through the pull of each slide fastener. Fold the webbing in the center, and bartack at the ends and close to the pulls.	Bartack		21-23 stitches per bartack
	(g) Cut four, 2-3/4 inches square of the basic fabric to reinforce the slide fasteners. Turn the outside edges of each reinforcement under 1/4 inch on all four sides, and stitch two reinforcements which will be adjacent to each end of the metal portion of the slide fastener (centered from the top and bottom on the inside of the pocket). "X" stitch inside box through all thicknesses of leather, fabric, and zipper tape.	301	LSa-1	10-12
15	Make Leg Socket Tab Assembly			
	a) Cut two pieces of nylon tape in accordance with the patterns, fold at the center, sew box "X"	301	OSf-1	10-12

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
15	3/4 x 1-1/4 inch, 1/8 inch from the folded edges, punch a hole 1/8 inch diameter through the center of box "X". Install button NASM27980-1B and socket NASM27980-6B through the hole at the center of box "X" stitch.			
16	Make Body Socket Tab Assemblies			
	(a) Cut two pieces of nylon tape in accordance with the patterns, fold at the center, sew a box "X" 3/4 x 1-1/4 inch, 1/8 inch from the folded edge, punch a hole 1/8 inch diameter through the center of box "X" stitch, install button NASM27980-1B and socket NASM27980-6B through the hole at the center of box "X" stitch.	301	OSf-1	10-12
17	Construct Bladder			
	(a) Construct the bladder as specified in 3.6.2.			
18	Make Body Section, Addition of Blank Label to Rear Abdomen Section			
	(a) Join the left side of the body front to the left side of the body back with a double row of stitching, 1/16 inch from the edges, 1/4-inch gage.	301	(see figure 4)	8-10

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
18	(b) Sew a 1-1/8-inch-wide, bias-cut piece of the basic fabric over the center of the body front and back body seam (inside) with a single row of stitching, 1/16 inch from the edge, each side of the binding.	301	LSd-1	8-10
	(c) Center the bias-cut pieces of the basic fabric on the stiffeners. Sew three rows of stitching from the end to end with one row in the center and one row on the right and the left sides of the center stitching 1/8 inch.	301	SSa-1	10-12
	Place the stiffeners at the patterns marks on the back body and the front body and stitch a single row of stitching, 1/16 inch from the edges, each side of the binding.	301	LSa-1	10-12
	(d) Place the four lacer loop tapes on the body back, in accordance with the pattern marks, the ends even with the top and bottom edges of the body back, loops facing loops, and join with a double row of stitching 1/16 inch from the edge, 3/16-inch gage.	301	LSa-1	10-12
	Bartack both ends of each lacer loop, across the full width of the lacer loop including the loop material, 3/8-inch from the top and bottom edges.	Bartack		21-22

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
18	(e) Place the pile fastener tape on the lacer loop tape on the body back, 1/8 inch from the loops, and join with a single row of stitching 1/16 inch from the edge.	301	SSa-1	10-12
	(f) Bind across the top, catching the seam tape and the lacer loop tape in the binding, with the 5/16-inch-wide (finished) bias binding with a single row of stitching 1/16 ± 1/32 inch from the edge.	301	BSc-1	10-12
	(g) Bind across the bottom of the body front and body back with a 5/16-inch-wide (finished) bias tape with a single row of stitching 1/16 ± 1/32 inch from the edge.	301	BSc-1	10-12
	(h) Place the size label either vertically or horizontally on the inside of the center back 2 inches below the lower edge of the binding and centered between the two stiffener seams, and sew with a single row of stitching 1/8 ± 1/16 inch from the edge on all four sides. Place the contractor's label below the size label on the center back, and sew with a single row of stitching 1/8 ± 1/16 inch from the edge on all four sides.	301	SSa-1	10-12

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
18	(i) Place the blank label (5-inch-long x 1-1/2-inches-wide) just below the lower edge of the binding on the remaining center back section between and parallel to the stiffeners. Sew label to back section with a single row of stitching 1/8 ± 1/16 inch from the edge on all four sides.	301	SSa-1	10-12
	(j) Place the instruction label below the blank label on the center back, and sew with a single row of stitching 1/8 ± 1/16 inch from the edge on all four sides.			
	(k) Cut a 4-inch length of the 5/16-inch-wide webbing. Bartack one end 3/4 inch down from the top of the edge binding-on-binding strip over the seam joining the body front to the side back. Fold the webbing to the inside; bartack the other end 1-3/4 inches from the top edge binding.	Bartack		21-23 Stitches per bartack
	(l) Turn the body lacer cover under 3/8 inch. Place the finished edge of the body lacer cover on the inner row of lacer loop tape, edge to edge; join with a double row of stitching 1/16 inch from the edge, 3/16-inch gage.	301	LSa-2	10-12

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
19	Attach Inflation Tube Opening Reinforcement			
	(a) Lay the 3-5/16-inch-square reinforcement piece of the basic fabric at the marked location on the outside of the body front, evenly spaced over the hole position. Sew around the hole marking, approximately 1-1/8-inch diameter, with a single row of stitching.	301	SSe-2(a)	10-12
	(b) Trim the 1/4 inch inside stitching; and make approximately six small cuts around the edge of the hole but no closer than 1/16 inch from the stitching. Turn the reinforcement piece through the opening; topstitch with a single row of stitching 1/16 inch from the topstitched edge.	301	SSe-2(b)	10-12
	(c) Turn the outside edge of the reinforcement under 1/2 inch all around; sew to the body front with a single row of stitching 1/16 inch from the folded edge.	301	LSd-1	10-12
	(d) Lay the octagon-shaped leather reinforcement piece evenly spaced over the hole position, on the inside of the body front, and sew around the hole 3/32 inch from the edge on the inside. Sew around the outside edge of the	301	LSa-2	9-11

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
19	leather patch 3/32 inch from the edge on the inside.			
	(e) Reinforce the inner face of the inflation sleeve external covering (see 3.4.1.1) as indicated by patterns by stitching down a 4-1/2 inch x 1-1/2 inch piece of coated fabric (see 3.4.1.2) 1/8 inch from the edge. On the outer face of the external tube covering over the reinforcement patch, stitch a 1 inch wide x 5 inch long piece of hook fastener tape 1/8 inch from the edge.	301	LSa-2	9-11
20	Attach Body Bladder Cover			
	(a) Place the assembled body bladder cover on the body front to the inside at the marked location with the bladder inflation sleeve on the left side; join with a double row of stitching 1/16 inch from the edge, 1/16-inch gage. Leave open the end of external inflation sleeve covering, the leg openings, and 1 inch at the tab location, top, bottom and right side.	301	SSa-2	10-12
21	Attach Assembled Body Protective Fly			
	(a) Turn the edge of the cloth under 3/8 inch on the body front. Sew the slide fastener tape side	301	LSd-2	10-12

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
21	with the protective fly attached, with 1/4 inch of the slide fastener tape exposed, with a double row of stitching 1/16 inch from the folded edge, 1/4 inch gage. The slide fastener shall be on the inside with the opening on the top and the closed end extending below the body panel by 1 ± 1/8 inch.			
	(b) Sew the upper body button/socket tab to the inside of the side-back (front) on the right-hand side of the slide fastener edge as marked on the patterns, forming a box and "X" stitch, 3/8 inch by 1-3/8 inches, 1/16 inch from the edge. The upper button/socket tab shall match the stud/ eyelet on the body protective fly. Sew the lower button/socket tab 5-1/2 inches (center line to center line) below the upper button/socket tab. The lower button/socket tab shall match the lower stud/eyelet on the body protective fly.	301	LSd-1	10-12
22	Make Leg Sections			
	(a) Sew a 5/8-inch-wide, bias-cut piece of the basic fabric on the inside of the legs, at the positions marked at the sides of the pocket and at the leg take-ups, with a single row of stitching 1/16 inch from the edge each side of the binding, right and left.	301	LSd-1	10-12

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
22	(b) Turn the neck of the pockets outside 1/4 inch; sew across the knees at the marked locations with a double row of stitching 1/16 inch from the edge 1/16-inch gage, leaving 1/2 inch at each edge open, the pocket to the outside, the slide fastener closing to the outside.	301	LSd-2	10-12
	(c) Sew a 2-1/2 inch wide, 4-1/2 inch long reinforcement made of the bladder material, on the inside of the left leg at the position marked for the pile fastener tape, with a single row of stitching 1/8 inch from the edge.	301	LSa-1	10-12
	(d) Sew the pile fastener tape on the outside of the left leg, at the marked position, with a single row of stitching 1/8 inch from the edge on all four sides.	301	SSa-1	10-12
	(e) Place the webbing, with the button/socket assembly NASM27980-1B, -6B attached to the webbing on the inside of the legs. The top edge of the webbing shall be parallel with and down from the top 1-3/4 inches. Sew across the back end of the webbing, following the contour of the bladder cover locations, with a single row of	301	LSd-1	10-12

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
22	stitching. The socket of the button/socket tab shall face the body and shall match the stud on the leg protective fly.			
23	Make and Attach Knife Pocket			
	(a) Lay the knife pocket reinforcement piece on the pocket fabric, face up, in accordance with the notches and the drill marks. Sew the knife pocket reinforcement piece to the pocket with a single row of stitching 1/16 to 1/8 inch from the edge on all four sides.	301	SSa-1	10-12
	(b) Fold the pocket fabric lengthwise with the raw edges even. Bind the square end with a 5/16-inch-wide (finished) bias binding with a single row of stitching 1/16 ± 1/32 inch from the edge. The leather reinforcement piece shall be on the outside after folding.	301	BSc-1	10-12
	(c) Bind the straight edge of the small flap reinforcement piece for the top of the pocket with 5/16 inch wide (finished) bias binding with a single row of stitching 1/16 ± 1/32 inch from the edge.	301	BSc-1	10-12

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
23	(d) Position the flap reinforcement piece in accordance with the notch and the drill marks at the round end of the pocket with the flap piece on the same side as the webbing piece. Sew the flap piece to the two top plies of the pocket fabric with a single row of stitching 1/8 ± 1/16) inch from the edge.	301	SSa-1	10-12
	(e) Place the leather reinforcement piece on the outside of the knife pocket, flat and smooth, with the bottom edge resting on the center notch on the knife pocket and the top edge toward the round end of the knife pocket. Stitch the leather reinforcement piece 1/8 inch ± 1/16 inch from the edge on all four sides. Double stitch across the top of the leather reinforcement piece with 1/4 inch gage.	301	SSa-1	10-12
	(f) Fold the pocket fabric on the center notch. Bind the pocket with 5/16 inch wide (finished) bias binding, starting at the bottom folded end and sewing the binding around the sides and the top with a single row of stitching 1/16 ± 1/32 inch from the edge. The leather reinforcement piece shall be on the outside after binding. The tape shall extend 1/2 inch beyond the end on each side for turning under.	301	BSc-1	10-12

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
23	(g) Place a single row of stitching across the flap through both plies at the round end of the flap, approximately 1 inch from the open bound end.	301	SSa-1	10-12
	(h) Install a size 00 grommet on the pocket, between the square end and the round end of the flap reinforcement piece, in accordance with the drill marks.	301	SSa-1	10-12
	(i) Position the reinforcement patch on the left thigh front so that the knife pocket is just inside the bladder case stitch line at the front, bottom and top corners, and so that the side of the pocket is parallel to the leg opening slide fastener. Turn the raw edges under 3/8 inch at the bottom and the sides; join with a double row of stitching 1/16 inch from the folded edge, 1/4 inch gage. Trim the top even with the top of the thigh.	301	LSd-2	10-12
	(j) Position the knife pocket on the reinforcement patch, with the round end up and the opening on the inside, in accordance with the drill marks. Start at the bound opening and sew down the side, across the bottom, and up to the bound opening, turning the bias end of the tape under on each side and backstitching 1/2 inch on each side,	301	LSd-2	10-12

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
23	with a double row of stitching 1/16 inch from the edge, 1/8 inch gage. The distance between the inside longitudinal stitch lines shall be 2-1/4 ± 1/8 inch. (k) Sew across the bound square end at the top of the pocket opening, securely tacking the top of the pocket to the patch and the Anti-G Garment, with a single row of stitching. (l) Stamp one snap fastener socket and button to the center of the flap so that the dot on the snap fastener will be at the top of the flap in accordance with the drill marks at the rounded end. (m) Stamp one snap fastener stud and eyelet, reinforced with bladder material on the inside of the Anti-G Garment to correspond to the snap fastener socket. If the stud and eyelet location is on top of the binding around the thigh opening, relocate position just below the binding. (n) Thread a 64-inch-long lanyard through the grommet. Tie the lanyard with a bowline knot. Sear both ends of the lanyard. Fold the lanyard in hanks that are approximately 4 inches long.	301	SSa-1	10-12

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
23	Position the folded lanyard in the pocket, parallel with the pocket, with all of the folds within the pocket. Close the snap fastener.			
	(o) Sew the pile fastener tape for the checklist fastener to the outside of the left leg, at the location shown on figure 5 with a single row of stitching 1/8 inch from the edge on all four sides.	301	SSa-1	10-12
	(p) Using figure 5 as a guide, sew the pile fastener tape for the checklist fastener to the right leg with a single row of stitching 1/8 inch from the edge of all four sides.	301	SSa-1	10-12
24	Attach Leg Bladder Covers			
	(a) Place the leg bladder covers on the legs at the marked locations on the patterns. Join down each side, leaving the top and 1 inch at the marked position for the tabs open, with a double row of stitching 1/16 inch from the edge 1/16-inch gage. Fold back 1/4 inch at the neck of the pocket left loose so that the pocket is not stitched through in this operation.	301	LSd-2	10-12
25	Attach Pockets			

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
25	(a) Sew the pockets to the lower legs on the center of the basic fabric bias binding, down each side, with a single row of stitching 1/16 inch from the edge of the pocket binding.	301	LSd-1	10-12
	(b) Join the pockets at the bottom and the upper edges at the knee opening to the legs with a single row of stitching 1/8 inch from the edge.	301	SSa-1	10-12
26	Bind Knee Openings and Bottom of Legs			
	(a) Bind the knee openings (including the pockets) and the bottom of the legs with 5/16-inch-wide (finished) bias binding with a single row of stitching 1/16 ± 1/32) inch from the edge.	301	BSc-1	10-12
27	Join Lacer Loop Tapes to Thighs and Calves			
	(a) Place the eight lacer loop tapes the length of each thigh and each calf, in accordance with the marks on the patterns, with the ends even with the top and the bottom edge, with loops facing loops. Join the lacer loop tapes to each thigh and each calf with a double row of stitching 1/16 inch from the edge, 3/16 inch gage.	301	SSa-2	9-11

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
27	(b) Bartack both ends of lacer loop, across the full with of the lacer loop including the stitches per loop material, 3/8 inch from the top and the bartack bottom edges.	Bartack		21-22
28	Join Pile Fastener Tape to Thighs and Calves			
	(a) Place the pile fastener tape on the lacer loop tape on each thigh and each calf, 1/8 inch from the loops, and join with a single row of stitching 1/16 inch from the edge.	301	SSa-1	10-12
29	Attach Assembled Leg (Thigh and Calf) Lacer Covers			
	(a) Turn the raw edge of each assembled leg (thigh and calf) lacer cover under 3/8 inch; place the finished edge of each leg lacer cover on the inner row of the lacer loop tape, edge to edge. Join with a double row of stitching 1/16 inch from the edge, 3/16 inch gage.	301	LSa-2	10-12
30	Join Thigh Take-Ups			
	(a) Join the assembled right and left take-ups to the outside at the thigh, in accordance with the pattern marks, down the sides and across the top, with a double row of stitching 1/16 inch from the	301	LSa-2	10

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
30	edge, 3/16 inch gage. The slide fasteners shall close at the top. Note: Use proper thread tension and fabric tension to avoid waviness (chain flatness of slide fastener).			
31	Sew Around Reinforcement			
	(a) Sew around the edges of the reinforcement inside of the leg take-ups with a single row of stitching 1/16 inch from the edge.	301	SSa-1	10-12
32	Attach Assembled Leg Protective Flies			
	(a) Turn the edge of the cloth and elastic subassembly under 3/8 inch. Place the edge of the cloth down the legs on the slide fastener tapes so that the separating ends of the slide fastener will be at the top, the protective fly will open to the back, the corners at the knee holes will be spaced 3/4 $\pm$ 1/8 inch apart on the slide fastener tape and will be exposed. Sew each fly to each slide fastener tape with a double row of stitching 1/16 inch from the edge, 1/4 inch gage.	301	LSd-2	10-12

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
33	Install Snap Fasteners (a) Install the snap fastener cap and socket through the lower edge of the legs to the side opposite the protective fly, 7/8 inch from the bottom edge and 5/8 inch from the edge of the slide fastener teeth edge to the center of the snap fastener, with the cap on the face side of the cloth. (b) Install the snap fastener stud and post, to match the snap fastener cap and socket, through the protective flies at the bottom of the legs, approximately 5/8 inch from the edge of the protective fly and 1 inch up from the bottom edge of the legs to the center of the snap fasteners, with the stud to the outside.			
34	Join Upper Ends of Legs to Body Front			
	(a) Join the upper ends of the legs to the body front with the face sides of the material together, in accordance with the marks on the patterns, and with a single row of stitching 3/8 inch from the edge.	301	LSq-2(a)	10-12
	(b) Turn and topstitch with a single row of stitching 1/16 inch from the folded edge.	301	LSq-2(b)	10-12

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
34	(c) Sew an additional row of stitching 5/16 inch from the folded edge.	301	SSa-1	10-12
35	Insert Adjustment Laces in Take-Ups (a) Select the adjustment laces in accordance with table V. (b) Insert the adjustment laces into the lacer loops for the body take-ups as follows: Start lacing from the bottom similar to lacing shoes. Route the adjustment laces through each lacer loop on each side (loose ends exiting from the top lacer loops). Leave the lacing loose and flat so that adjustment is open to maximum width.. Fold the excess lacing into a hank that is approximately 3 inches long. Wrap the hank securely with a rubber band. (c) Insert the adjustment laces into the lacer loops for the thigh take-ups in the same manner as specified for the body take-ups in No. 35(b), except that the lacing shall start from the top lacer loops and exit from the bottom lacer loops. Fold the excess lacing into a hank that is approximately 3 inches long. Wrap the hank securely with a rubber band.			

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
35	(d) Insert the adjustment laces into the lacer loops for the calf take-ups in the same manner as specified for the body take-ups (see No. 35(b)).			
36	Final Assembly			
	(a) Install the bladder through the openings at the top of the legs left open for this purpose and into the bladder covers. Insert the bladder tabs through the 1 inch wide openings; sew with a double row of stitching 1/16 inch from the edge, 1-1/16 inch gage.	301	SSa-2	10-12
37	Make Connector Assembly			
	(a) See section U-U of figure 1. Insert the free end of the rubber sleeve (see 3.4.17) the unfolded end of the mesh spacer tube (the end with section A-A of Drawing 1628AS900), and insert interlaced springs (see 3.4.20) into free end of rubber sleeve. Lay one end of the spacer fabric (see 3.4.3) over the spacer tube so that it ends 1/4 inch from the wide labeled part of the connector. Connect the spacer tube and spacer fabric to the rubber sleeve by clamping in place with NAS397-14, 1 inch from the wide labeled part of the connector. Wrap the clamped area with at			

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
37	least two full turns of wrapping tape. Clamped. areas shall be centered within the one inch width of the wrapping tape			
38	Attach Connector Assembly to Inflation Tube (a) Make an insertion tool from a coat hanger or any other wire by clipping and filing the end round and smooth. (b) Insert the insertion tool into the folded end of the spacer tube (see C-C of Drawing 1628AS900). Push the spacer assembly into the bladder inflation hose, working the spacer tube in slowly so that it lies adjacent to the existing Trilok spacer and so that the spacer extends approximately 4 inches into the body bladder. When the spacer tube is completely inserted, hold on to the folded end of the spacer tube from the outside of the body bladder and pull the insertion tool out. (c) See section U-U of figure 1. Pull the bladder inflation tube over the spacer tube assembly and rubber sleeve flush with the wide labeled part of the connector. Clamp the bladder inflation tube to the rubber sleeve of the connector assembly with a NAS397-16 clamp, clamp toward connector.			

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TABLE III. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
38	Wrap the clamped area with at least two full turns of wrapping tape. Clamped areas shall be centered within the one inch width of the wrapping tape. (d) See section U-U of figure 1. Pull external inflation tube covering (see 3.4.1.1) over the bladder tube up to the wide labeled part of the connector. Clamp in place with NAS397-16 clamp, clamp toward wide labeled part of the connector. Wrap the clamped area with at least two full turns of wrapping tape. Clamped areas shall be centered within the one inch width of the wrapping tape. (e) See section U-U of figure 1. Install the rubber band (see 3.4.22) over all the wrapped, clamped areas so that one third of the rubber band covers the wide labeled part of the connector and the rest covers the wrapped clamped area. All wrapping tape shall be covered by the rubber band.			

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TABLE IV. Dimensions of fastener tapes.

Size of Anti-G Garment	Dimensions of Fastener Tape (Inches)				
	Body Lacer Cover 1/ 2/	Thigh Lacer Covers 1/ 2/	Hose Retainers 3/	Checklist Retainers 4/	Calf Lacer Covers 1/ 2/
Small Regular	1 by 9-1/2	1 by 10-5/8	4	2 by 4	1 by 12-1/2
Small Long	1 by 9-1/2	1 by 11-1/2	4	2 by 4	1 by 13-1/8
Medium Regular	1 by 10	1 by 10-7/8	4	2 by 4	1 by 12-1/2
Medium Long	1 by 10	1 by 11-3/8	4	2 by 4	1 by 13-1/8
Large Regular	1 by 10	1 by 10-7/8	4	2 by 4	1 by 12-1/2
Large Long	1 by 10	1 by 11-1/4	4	2 by 4	1 by 13-1/4
Large Extra Long	1 by 10	1 by 11-1/4	4	2 by 4	1 by 17-1/4

1/ Since the requirements of table III specify that the fastener tape for the lacer covers shall be trimmed, if necessary, so that the fastener tape will be even with the top and the bottom edges of the flap, these dimensions are approximate.
2/ Two for each Anti-G Garment shall be required.
3/ One for each Anti-G Garment shall be required. The width shall be 1 inch for the hook fastener tape and 2 inches for the pile fastener tape.
4/ Two of the pile fastener tape shall be required.

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TABLE V. Adjustment laces.

Size of Anti-G Garment	Length of Adjustment Laces (Feet) 1/		
	Waist Area 2/	Thigh Area 2/	Calf Area 2/
Small Regular	12	12	14
Small Long	12	13	16
Medium Regular	12	13	16
Medium Long	12	14	17
Large Regular	13	14	17
Large Long	13	16	19
Large Extra Long	13	16	19

1/ The dimensions specified in this table are approximate and shall be governed by the following requirement: Both ends of each adjustment lace shall extend at least 6 inches beyond the open end of the lacing tape when the adjustment laces are completely extended. Both ends of each adjustment lace shall be dipped in a hot solution of 50 percent beeswax and 50 percent paraffin.

2/ Two per Anti-G Garment shall be required.

3.6.2 Bladder. The bladder shall be constructed as follows: The polyurethane coated nylon taffeta cloth specified for the bladder in 3.4.1.2 shall be cut in accordance with the applicable patterns. The bottom half of the bladder with the coated surface face up shall be laid on a clear flat surface. Four strips of the spacer material specified in 3.4.3 shall be cut in lengths required by the applicable pattern except that the horizontal spacer extending across the inside of the abdominal bladder shall have approximately 2 to 3 inches of slack (the length of spacer between the two patches in the abdominal bladder shall be 2 to 3 inches longer than the distance in the flat between the two patches). The width of the spacer in the bladder extension and the abdominal bladder shall be 3/4 inch. The width of the leg spaces shall be 1 inch. The ends of the legs and abdominal spacer material shall be overlapped and joined to patches that are made of the polyurethane coated nylon cloth specified in 3.4.1.2, and that are approximately 1 inch by 1-3/4 inches by bartacking as shown in figures 2 and 3. Two additional pieces of spacer material 1 inch wide by 10 inches long shall be sewn to the upper portion of the leg spacer material (one length per leg). The upper end of the piece of spacer material that is 1 inch by 10 inches shall be placed approximately 1 inch below the patch joining the leg spacer to the abdominal spacer and shall be sewn to the leg spacer with two rows of stitching 1/4 inch from each edge of the leg

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spacer with the thread specified in 3.4.11, seam ASTM-D6193, Type LSa-2, 8 stitches per inch, and all ends backstitched a minimum of 1 inch. The bladder extension spacer material shall be overlapped on the abdominal spacer material approximately 2-1/2 inches from the joint between the leg and abdominal spacer material and shall be similarly bartacked (no patch). The spacer material (anti-kink system) shall be positioned on the coated side of the bottom half of the bladder as indicated on the applicable pattern and then secured by cementing patches to the bottom of the bladder as shown in figure 2. The top half of the bladder shall be superimposed and aligned on the bottom half of the bladder. The bladder shall be positioned welded together so that the bonded area will be 1/8 $\pm$ 1/32 inch wide, 3/8 $\pm$ 1/16 inch from the edge of the bladder, and 1/16 inch from each seam. The welded seam shall be straight, continuous, and parallel to the edge of the bladder. No part of the spacer material or of the attachment patches shall be caught in any portion of the welded seam. The completed bladder shall then be inflated to 2 pounds per square inch (psi) air pressure. The seams of the inflated bladder shall be inspected for proper joining and adhesion. The inflatable area of the completed bladder shall be not greater than 5/8 inch nor less than 1/4 inch from the edge of the patterns. The upper tab of the finished bladder shall be stamped for identification with the abbreviated Anti-G Garment size.

3.7 Performance.

3.7.1 Leakage. The bladder shall not lose more than 0.5 psig of air pressure in 60 seconds when tested as specified (see 4.6.2).

3.7.2 Endurance. The Anti-G Garment shall not develop any structural defects when tested as specified (see 4.6.3). Structural defects shall include, but not be limited to, material torn, seam separation, slide fastener slider lock broken, slide fastener chain separated, snap fasteners unsnapped, clamps loosened or broken. After the Anti-G Garment has met the requirements of 3.7.2, the Anti-G Garment shall meet the requirements in 3.7.1.

3.7.3 Inflation time. The Anti-G Garment shall inflate to a pressure of 8 psig in not more than 3 seconds when tested as specified (see 4.6.4).

3.7.4 Low temperature storage. The Anti-G Garment shall not be adversely affected by storage at a temperature of -60 °F when tested as specified (see 4.6.5).

3.7.5 High temperature storage. The Anti-G Garment shall not be adversely affected by storage at a temperature of 160 °F when tested as specified (see 4.6.6).

3.7.6 Low temperature operation. The Anti-G Garment shall operate when tested at a temperature of -30 °F when tested as specified (see 4.6.7).

3.7.7 Bond strength (bladder). The force required to separate the bonded bladder halves shall be not less than 40 pounds when tested as specified (see 4.6.8).

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3.8 Finished measurements. (Refer to table VI and figure 6) The finished measurements of the Anti-G Garment shall be as specified in table VI. The minimum finished measurements specified in table VI shall be the measurements of the finished Anti-G Garment when its adjustment laces are tight. The maximum finished measurements specified in table VI shall be the measurements of the finished Anti-G Garment when its adjustment laces are loose and the fabric is stretched to the limit of the solid fabric with the lacer cover closed.

3.9 Color. The color of the materials of the Anti-G Garment shall be as specified herein. Materials for which color requirements have not been specified and which are hidden by the construction may be used in natural or colored form. The color of all exposed sewing threads shall approximately match the color of the basic fabric specified (see 3.4.1.1).

3.10 Identification of product.

3.10.1 Labels. Each Anti-G Garment shall have a contractor's label, a size label, and a laundering instruction label.

3.10.1.1 Contractor's label. The contractor's label shall conform to MIL-DTL-32075, Type VI, Class 1 and shall be approximately 2-1/2 by 1-1/2 inches. The label shall contain the following information:

Anti-G Garment, Cutaway, CSU-15A/P
Specification 1/
Manufacturer's Identification 1/
Contract or Order No. 1/

1/ The manufacturer shall insert the applicable information.

3.10.1.2 Size label. The size label shall conform to MIL-DTL-32075, Type VI, Class 2 and shall be approximately 2-1/2 by 1-1/2 inches. In addition to the adjective size, the size label shall contain the stature and the weight ranges (see 6.4) and the stock number of the Anti-G Garment in the following manner:

Size 1/
Stature 1/ Weight 1/
Stock No. 1/

1/ The manufacturer shall insert the applicable information.

3.10.1.3 Blank label. The blank label shall conform to MIL-DTL-32075, Type I, Class 2 and shall be approximately 5 by 1-1/2 inches.

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TABLE VI. Finished measurements.

Size of Anti-G Garment	Finished Measurements in Inches								
	Waist Circumference 1/		Thigh Circumference 2/		Leg Length at Fastener Location 3/		Ankle Circumference		Hose Length
	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum 4/	Maximum 5/	Length 6/
Small Regular	29-1/2	34-1/2	21	26	26	26	12	15	17
Small Long	29	34-1/4	20-3/4	25-1/2	28	28	12	15	17
Medium Regular	32-1/4	38-1/4	22-1/4	27-1/4	26-1/2	26-1/2	12	15	17
Medium Long	32	38	22-1/2	27-1/2	28-1/2	28-1/2	12	15	17
Large Regular	35-1/4	41	23-1/2	28-3/4	27	27	12	15	17
Large Long	35-1/2	41-1/4	24	29	29	29	12	15	17
Large Extra Long	35-1/2	41-1/4	24	29	33	33	13-3/4	16-3/4	17

1/ The tolerance shall be + 3/4 or - 1/2 inch.
2/ The tolerance shall be + 1/2 or - 3/8 inch.
3/ The tolerance shall be ± 1/2 inch.
4/ The tolerance shall be + 0 or - 1/2 inch.
5/ The tolerance shall be ± 1/4 inch.
6/ The tolerance shall be ± 1/4 inch.

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3.10.1.4 Laundering instruction label. The laundering instruction label shall conform to MIL-DTL-32075, Type VI, Class 3 and shall be approximately 4 inches by 1-1/2 inches. The instruction label shall contain the following information:

DO NOT MACHINE WASH, MACHINE DRY, IRON
OR DRY CLEAN TO AVOID POTENTIAL DAMAGE
TO QUICK DISCONNECT HARDWARE AND HOSE
ASSEMBLY.
AS NEEDED WASH BY HAND WITH MILD SOAP
AND COLD WATER. PLUG AIR INLET PORT
SECURELY TO PREVENT WATER FROM
ENTERING BLADDERS.

THE MATERIAL OF THE OUTER SHELL OF THIS
ANTI-G GARMENT IS AN INHERENTLY FIRE
RESISTANT MATERIAL THAT WILL NOT LOSE
ITS FIRE RESISTANT PROPERTIES DURING
WASHING.

3.10.1.5 Barcode label. Each item shall be individually barcoded with MIL-DTL-32075, Type VIII, Class 17, label/tag. The label/tag shall be located on the bag so that it is completely visible when it is folded or packaged as specified.

3.10.2 Special identification. The abdominal area of each Anti-G Garment that has been subjected to the endurance test specified in 4.6.3 shall be marked, with waterproof ink, in letters that are at least 1 inch high. The marking shall be as follows: ENDURANCE TESTED – NOT TO BE USED IN FLIGHT. The shipping containers for Anti-G Garments that have been subjected to the endurance test specified in 4.6.3 shall be marked in letters that are at least 1 inch high that repeats ENDURANCE TESTED – NOT TO BE USED IN FLIGHT.

3.11 Patterns. The Government will provide a complete set of patterns (see 6.2) which show size, directional lines, placement marks, and notches for assembly. The Government pattern shall be used to create a working pattern. The government pattern shall not be altered. Minor modifications are permitted to the working pattern where necessary when using automatic equipment or to accommodate a manufacturing process. These modifications shall not alter the dimensional, serviceability, or appearance requirements cited in this specification.

3.11.1 Directional line. A straight line with an arrow at each end and the work straight affixed to the straight line marked on a pattern part indicates the straight of the material which is also defined as the warp or lengthwise direction of the cloth or fabric. The pattern part shall be laid on the cloth so that the straight line with the arrows indicating the straight of the fabric is parallel to the selvage edge (lengthwise) of the fabric. This shall be accomplished by placing the straight line on the lengthwise grain of the fabric as follows: Measure the distance, at two points,

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from the straight line to the selvage. Adjust the pattern part until the distance from the straight line to the selvage is the same from both points (see figure 7).

3.12 Government loaned property. A connector conforming to MS27755 to be used to test the Anti-G Garment will be loaned, upon request, by the Government (see 4.2.2 and 6.2).

3.13 Workmanship. Completed garments shall be free of all loose thread, lint, foreign matter, irregularities or defects that could adversely affect performance or durability.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements of the Anti-G Garment are specified herein and classified as follows:

(a) Qualification inspection (see 4.3)
(b) First article inspection (see 4.4)
(c) Conformance inspection (see 4.5)

4.2. Inspection conditions.

4.2.1 Atmospheric conditions. Unless otherwise specified herein or in specified test methods, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury, at a temperature of 77 $\pm$ 5 °F, and at a relative humidity of 80 percent or less. Where tests are made with atmospheric pressure or temperature substantially different from these values, allowance shall be made for the change in instrument reading.

4.2.2 Connector. A connector conforming to MS27755 (see 3.12 and 6.2) shall be used to connect the Anti-G Garment to the air source for tests that require inflation and deflation cycling of the Anti-G Garment.

4.2.3 Air. The air used to test the Anti-G Garment as specified herein shall not contain any oil or moisture.

4.2.4 Air pressure gauge. Air pressure gauges utilized shall be in pounds per square inch (psig), and shall be precise to at least one hundredth of a psig. The gauge shall have a current calibration.

4.3 Qualification inspection. The qualification samples described in 4.3.1 shall be subjected to the qualification inspection. The qualification inspection for the two Anti-G Garments shall consist of the test methods specified in 4.6.1 through 4.6.7. The qualification inspection for the bladder shall consist of the test method specified in 4.6.8 (see table VII).

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TABLE VII. Qualification inspection.

Inspection	Requirement	Test Method
Visual Examination	3.5	4.6.1
Leakage	3.7.1	4.6.2
Endurance	3.7.2	4.6.3
Inflation Time	3.7.3	4.6.4
Low Temperature Storage	3.7.4	4.6.5
High Temperature Storage	3.7.5	4.6.6
Low Temperature Operation	3.7.6	4.6.6
Bond Strength (Bladder)	3.7.7	4.6.8
Dimensional/Finished Measurement	3.8 / table VI	figure 6

4.3.1 Qualification samples. The qualification samples shall consist of one Anti-G Garment in size large long, one Anti-G Garment in size medium regular, two complete bladders in size medium regular, and one complete bladder in size large long. (Note: Although the qualification samples include three bladders, only one bladder shall be subjected to the qualification inspection.)

4.4 First article inspection. The first article samples specified in 4.4.1.1 shall be subjected to the first article inspection. The first article inspection for the Anti-G Garment shall consist of the test methods specified in 4.6.1 through 4.6.7. The first article inspection for the bladder shall consist of the test method specified in 4.6.8 (see table VIII).

TABLE VIII. First article inspection.

Inspection	Requirement	Test Method
Visual Examination	3.5	4.6.1
Leakage	3.7.1	4.6.2
Endurance	3.7.2	4.6.3
Inflation Time	3.7.3	4.6.4
Low Temperature Storage	3.7.4	4.6.5
High Temperature Storage	3.7.5	4.6.6
Low Temperature Operation	3.7.6	4.6.7
Bond Strength (Bladder)	3.7.7	4.6.8
Dimensional/Finished Measurement	3.8 / table VI	figure 6

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4.4.1 First article samples. The first article samples shall be manufactured after the award of the contract and before production of the Anti-G Garment. The first article samples shall consist of the items specified in 4.4.1.1 and shall be representative of the materials, the components, the construction, and the workmanship to be used in the production items. If a manufacturer is in continuous production of the Anti-G Garments from contract to contract, inspection and submission of further first article samples may be waived at the discretion of the procuring activity (see 6.2). Approval of the first article samples or the waiver of the first article inspection shall not preclude the requirements for the performance of the conformance inspection.

4.4.1.1 First article sample quantities. The manufacturer of the Anti-G Garments shall furnish one complete Anti-G Garment of the following sizes: Small Regular, Medium Long, Large Regular, and Large Extra Long. The manufacturer of the Anti-G Garments shall furnish one complete bladder assembly of the following sizes: Small Regular, Medium Long, Large Regular and Large Extra Long.

4.5 Conformance inspection. The conformance inspection shall be performed on the completed Anti-G Garment and bladder assemblies (see 4.5.3 through 4.5.3.3). Unless otherwise specified within this specification, the verifications shall be performed in accordance with ANSI/ASQC Z1.4. Each lot shall be inspected for conformance. Each lot inspection shall consist of the examinations and tests specified in table IX. The presence of any nonconformance to table IX shall be cause to reject any production lot.

TABLE IX. Conformance inspection.

Inspection	Requirement	Test Method
Visual Examination	3.5	4.6.1
Leakage	3.7.1	4.6.2
Endurance	3.7.2	4.6.3
Inflation Time	3.7.3	4.6.4
Dimensional/Finished Measurement	3.8 / table VI	figure 6

4.5.2 Sampling inspection. Sampling inspection shall be conducted in accordance with plans A and B for the end items and components (see 4.5.3.2 and 4.5.3.3).

4.5.3 Plans for end items and components.

4.5.3.1 Lot formation. Each lot for plan A shall consist of 500 Anti-G Garments that have been manufactured essentially under the same conditions and submitted for inspection at substantially the same time. Each lot for plan B shall consist of 501 bladders that have been manufactured essentially under the same conditions and submitted for inspection at substantially the same time. A fraction of a lot shall not be permitted for plan A unless the total quantity of

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the Anti-G Garments required to be delivered during a specified period under the contract is less than 500 Anti-G Garments.

4.5.3.2 Plan A. Two Anti-G Garments shall be selected at random from each lot of Anti-G Garments or fraction thereof provided a fraction of a lot is permitted. Each Anti-G Garment selected as a sample shall be subjected to the following test methods as specified under 4.6:

(a) Endurance
(b) Inflation Time

4.5.3.3 Plan B. One bladder shall be selected at random from each lot of bladders or fraction there of provided a fraction of a lot is permitted. The bladder selected as a sample shall be subjected to the test method specified in 4.6.8.

4.5.3.4 Rejection and retest. When one or more items selected from a lot in accordance with plan A or B fail to meet the specification, acceptance of all items in the lot shall be withheld until the extent and cause of the failure have been determined. The contractor shall explain fully to the Government representative and notify the procuring activity in writing the cause of the failure, the action taken to preclude recurrence, and the impact this failure may have in scheduled deliveries. After correction, all of the sampling tests shall be repeated.

4.6 Test methods.

4.6.1 Examination of product. The Anti-G Garment shall be examined to determine compliance with the requirements of this specification with respect to materials, finished measurements, identification marking, and workmanship. Defects found during this examination shall be classified as specified in table X.

TABLE X. Classification of defects.

Defect	Classification		
	Critical	Major	Minor
MATERIALS Materials not conforming to specified requirements unless otherwise classified herein	1		
FABRIC IMPERFECTIONS See FED-STD-4 1/		100	

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TABLE X. Classification of defects - Continued.

Defect	Classification		
	Critical	Major	Minor
SLIDE FASTENERS			
Fabric seam on tape catching in slider or slider not operating freely		101	
Slide fastener not specified type, style, or length	2		
Any part of slide fastener assembly omitted, bent, or broken	3		
Pull tabs missing		102	
Stiffeners missing	4		
Pull tabs not specified type		103	
Stiffeners not specified type		104	
COLOR			
Color not matching specified color shade		105	
Any component not specified color		106	
SNAP FASTENERS			
Stud not securely clinched, damaged spring, or damaged or missing snap fastener		107	
Stud not aligned with the socket causing noticeable bulge or twist when snapped		108	
Stud misaligned with socket		109	

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TABLE X. Classification of defects - Continued.

Defect	Classification		
	Critical	Major	Minor
Snap fastener not holding together		110	
Snap fasteners not separating under normal pull 2/		111	
ASSEMBLY			
Incorrectly assembled, i.e., reversed seams, not specified type of seam, not specified type of stitch, or misplaced or reversed components	5		
Incompletely assembled, i.e., any component or required operation omitted	6		
IDENTIFICATION OF PRODUCT			
Any label missing, misplaced, illegible, or information on label incorrect or incomplete		112	
WORKMANSHIP			
Holes, cuts, tears, or needle chews	7		
Darns or mends affecting serviceability	8		
Darns or mends affecting appearance		113	
Stains, oil spots, or dirt over 1/4 inch in diameter		114	
Stains, oil spots, or dirt up to 1/4 inch in diameter in not more than two places			200
Stains, oil spots, or dirt up to 1/4 inch in diameter in three or more places		115	
MEASUREMENTS			
Any finished measurement not within specified tolerance		116	

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TABLE X. Classification of defects - Continued.

Defect	Classification		
	Critical	Major	Minor
Width of inside opening of knife pocket (distance between longitudinal stitch lines) less than 1-3/4 inches.		117	
HOOK ASSEMBLY			
Hook assembly not specified type		118	
Hook not aligned with loop (tolerance 1/8 inch - 3/; greater than 1/8 inch but less than 1/4 inch; greater than 1/4 inch)		119	
STITCHES AND SEAMS			
Open Seam	9		
Stitching missing, broken, skipped, or cut	10		
Run-offs	11		
Stitching margins unless otherwise classified herein 4/ …			
-up to 1/16 inch stitching margins not within tolerance of + 1/32 or - 0 inches			201
-over 1/16 inch to 1/4 inch stitching margins (tolerance + 1/32 or - 0 inch) that are greater than the + tolerance or less than + 1/16 inch			202
-over 1/16 inch to 1/4 inch stitching margins (tolerance plus 1/32 or minus 0 inch) that are greater than the tolerance of more than + 1/16 inch or more than - 1/32 inch			203
-over 1/4 inch to 1/2 inch stitching margins (tolerance + 1/16 inch or - 1/32 inch) that are greater than the tolerance by more than + 1/16 inch or more than - 1/32 inch		120	

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TABLE X. Classification of defects - Continued.

Defect	Classification		
	Critical	Major	Minor
-over 1/4 inch to 1/2 inch stitching margins (tolerance + 1/16 inch or - 1/32 inch) that are greater than the tolerance of less than + 1/16 inch or less than - 1/32 inch			204
Seams not within tolerance of ± 1/32 inch …			
-up to 1/4-inch seams that are greater than the tolerance up to + 1/32 inch			205
-up to 1/4-inch seams that are greater than the tolerance of more than ± 1/32 inch		121	
-over 1/4-inch seams that are greater than the tolerance up to + 1/16 inch			206
-over 1/4-inch seams that are greater than the tolerance of more than + 1/16 inch or - 1/32 inch		122	

1/ Fabric imperfections that would affect the appearance or the function of the Anti-G Garment shall be considered major defects.

2/ The snap fastener shall be checked for proper function and attachment by snapping closed and unsnapping the snap fastener at least three times.

3/ The hook assembly shall be checked by attaching the hook to the loop and closing the slide fastener. Alignment of the hook assembly shall not interfere with the slide fastener operation.

4/ This classification of defects for stitching margins shall not apply to any stitching margins for which a tolerance has been specified in table III. A stitching margin that is not within the tolerance specified in table III shall be classified as a major defect.

4.6.2 Leakage. The bladder in the Anti-G Garment shall be inflated to a pressure of 12 psig as indicated by a gage. The air supply shall be shut off securely. After a minimum of 60 seconds the pressure in the bladder shall be checked and readjusted, if necessary, to the original pressure of 12 psig. A minimum of 60 seconds after the pressure has been checked and re-adjusted, the pressure in the bladder shall be checked. If the pressure in the bladder is less than 11.5 psig, the Anti-G Garment shall be rejected. While the Anti-G Garment is still inflated, it shall be examined for material and constructional failure, separation of seams and components, change in location of slide fastener sliders, separation of slide fastener chains, and twisting and distortion of the Anti-G Garment. After this examination is completed, the bladder of the Anti-G Garment shall be completely deflated.

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4.6.3 Endurance. The Anti-G Garment shall be fitted to the inanimate model specified in 4.8. The Anti-G Garment shall be inflated 1,000 times, in accordance with table XI, to a pressure of not less than 15 psi and then examined for structural defects. Upon completion of the endurance test the Anti-G Garment shall be leakage tested in accordance with 4.6.2.

TABLE XI. Inflation of Anti-G Garment.

Minimum Pressure (psi)	Maximum Seconds Permitted to Inflate Garment to Minimum Pressure	Minimum Seconds Permitted to Deflate Garment to Zero psi (Gage Pressure Reading)	Minimum Cycles
15	8	4	5 per minute
12	8	4	5 per minute
6	30	10	10 per hour

4.6.4 Inflation time. The Anti-G Garment shall be fitted to the inanimate model specified in 4.8. The Anti-G Garment shall be connected to a source of air with a free flow of 10 cubic feet per minute. The regulator shall be adjusted so that the pressure shall be not greater than 12 psig. The bladder in the Anti-G Garment shall inflate to a pressure of 8 psi in not greater than 3 seconds when the pressure is applied.

4.6.5 Low temperature storage. The Anti-G Garment shall be packaged in a corrugated box with maximum dimensions of 13-1/2 by 4-1/2 by 8-1/2 inches. The packaged Anti-G Garment shall be subjected to a temperature of -60 °F or colder for at least 48 hours. The packaged Anti-G Garment shall then be kept at room temperature for at least 24 hours. The Anti-G Garment shall be removed from the box, fitted to the inanimate model specified in 4.8, and inflated 500 times, in accordance with table XI, to a pressure of not less than 12 psi. At the completion of this test, the Anti-G Garment shall be examined for failures or defects in the material or workmanship.

4.6.6 High temperature storage. The Anti-G Garment shall be packaged in a corrugated box with maximum dimensions of 13-1/2 by 4-1/2 by 8-1/2 inches. The packaged Anti-G Garment shall be subjected to a temperature of +160 °F or warmer for at least 4 hours. The packaged Anti-G Garment shall then be kept at room temperature for at least 24 hours. The Anti-G Garment shall be removed from the box, fitted to the inanimate model specified in 4.8, and inflated 500 times, in accordance with table XI, to a pressure of not less than 12 psi. At the completion of this test, the Anti-G Garment shall be examined for failures or defects in the material or workmanship.

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4.6.7 Low temperature operation. The Anti-G Garment shall be subjected to a temperature of - 30 °F or colder for a period of 4 hours. During the 4-hour period the Anti-G Garment is being subjected to the - 30 °F, it shall be inflated at the rate of 10 times per hour to a pressure of not less than 6 psig in accordance with table XI. At the completion of this test, the Anti-G Garment shall be examined for failures or defects in the material or workmanship. Structural defects or any other defects shall be cause for rejection of the Anti-G Garment.

4.6.8 Bond strength (bladder). Five specimens shall be cut from different locations of the bladder so that the specimens will be representative of the sealing of the entire bladder. The specimens shall be taken from the straight of the bladder only. Each of the specimens shall be cut across and perpendicular to the seam of the bladder and shall be 1 $\pm$ 1/16 inch wide and 6 $\pm$ 1/16 inches long. A suitable inspection apparatus equipped with an autographic recording device (see ASTM-D5034) shall be used to conduct the bond strength test. The front clamp of each jaw of the inspection apparatus shall be 1 inch by 3 inches. The rate of separation of the jaws of the inspection apparatus under no load shall be 12 $\pm$ 1/2 inches per minute. One of the free ends of a specimen (an end that is not bonded) shall be placed in the upper jaw of the inspection apparatus. The other free end of the specimen shall be placed in the lower jaw of the inspection apparatus. The bonded portion (seam of bladder) of the specimen shall be in the center between the two jaws. The jaws shall be separated until either breakage of the cloth or separation of the bond occurs. The highest value required to separate the bonded specimens or to break the cloth shall be recorded as the test result of the individual bonded specimen. This test shall be repeated until the other four specimens have been tested for bond strength. The lowest value obtained from the tests of the five specimens shall be reported as the test result for the bond strength test and shall be not less than 40 pounds.

4.7 Finished measurements. The finished measurements (see figure 6) shall be taken with the garment spread flat. All measurements shall be taken utilizing a retractable metal tape measure. The finished measurements shall be in accordance with table VI. The following measurements will be taken:

a. Maximum waist circumference - With the waist adjustment laces fully opened, measure the waist circumference straight across the top edge of the waist from the slide fastener end stop to slide fastener end stop.

b. Minimum waist circumference - With the waist adjustment laces fully tightened, measure the waist circumference straight across the top edge of the waist from the slide fastener end stop to slide fastener end stop.

c. Maximum thigh circumference - With the thigh take up slide fasteners fully closed, and adjustment laces fully opened, measure from the slide fastener retainer unit of the rear thigh to slide fastener pin unit of the front thigh.

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d. Minimum thigh circumference - With the thigh take up slide fasteners fully closed, and adjustment laces fully tightened, measure from the slide fastener retainer unit of the rear thigh to slide fastener pin unit of the front thigh.

e. Leg length - Measure at the slide fastener location from the top of the slide fastener retainer to the bottom end stop of the slide fastener of the leg opening.

f. Maximum ankle circumference - With the calf adjustment laces fully loosened, measure straight across the leg below the calf pocket from the rear slide fastener top stop to the front slide fastener top stop.

g. Minimum ankle circumference – With the calf adjustment laces fully tightened, measure straight across the leg below the calf pocket from the rear slide fastener top stop to the front slide fastener top stop.

h. Sleeve assembly – Measure the length of the hose assembly from the end of the MIL-C-83390 connector (see figure 1) to the point where the hose sleeve inserts to the hole in the left body front.

4.8 Inanimate models. Two inanimate models, each consisting of two legs and one torso, shall be required. The Anti-G Garments in sizes small regular and small long shall be fitted to the small model. The Anti-G Garments in sizes medium regular, medium long, large regular, large long, and large extra long shall be fitted to the large model. The inanimate models shall conform to figure 8 and table XII. The inspecting activity will inspect the mannequin for surface smoothness prior to any test.

TABLE XII. Models.

Models	Leg Dimensions in Inches 1/				
	A	B (dia)	C (dia)	D	E
Small	33	6-3/4	2-11/16	1-1/4	1-1/4
Large	33	7-3/16	3-1/16	1-1/4	1-1/2

Models	Torso Dimensions in Inches 1/						
	F	G	H	I	J	K	L
Small	14-1/4	12-3/4	8-1/16	10-1/8	6-13/16	1-1/4	1-1/4
Large	14	14-3/4	9-1/8	11-3/8	7-15/16	1-1/4	1-1/2

1/ Tolerances on all dimensions shall be ± 1/16 inch.

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4.8.1 Fitting Anti-G Garment to models. Fitting of the Anti-G Garment for all tests that require the use of the inanimate models shall be accomplished as follows: Wooden or plastic spacers shall be inserted between the Anti-G Garment and the model. The spacer for the torso section shall be 4 inches wide by 12 to 14 inches long by 3/4 inch thick and shall be inserted lengthwise so that the 4-inch width will face the model. The spacer for each leg shall be 2 inches wide by 12 to 14 inches long by 3/4 inch thick and shall be inserted lengthwise so that the 2-inch width will face the model. The adjustment laces of the Anti-G Garment shall then be taken up and tied so that the Anti-G Garment will fit snugly, but not tightly, around the abdomen and thighs. All slide fasteners shall be in the fully closed position. The model shall be placed in a sitting position approximately that of a pilot seated in the cockpit of an aircraft with the seat back angle of approximately 15 degrees. If a snug fit cannot be obtained at the waist of the sizes large long and large extra long Anti-G Garments, shims shall be affixed to the model on both sides. All spacers shall be removed prior to commencement of the tests.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Anti-G Garment covered by this specification is intended to be worn by high performance aircraft aircrew to counteract G (gravity) forces.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification, including any amendments.

b. Size required (see 1.2).

c. Government furnished patterns (see 3.11).

d. Waiver or requirements for first article inspection, including first article test reports and shipping instruction (see 3.2 and 4.4 through 4.4.1).

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e. Government loaned connector (see 3.12 and 4.2.2).

f. Standard color sample (see 3.4.1.1).

g. Packaging requirements (see 5.1).

h. Destination of tested Anti-G Garments (see 6.5).

6.3 Information requests. For access to information such as patterns, drawings, standard samples, etc. visit DSCP's Warfighter website, http://warfighter.dla.mil. Choose Contracting/ Specification /Pattern Request. Complete the request form, then submit.

6.4 Stature and weight ranges. Proper fitting of the Anti-G Garment is based on stature and weight; other body measurements are not used in selecting the proper size. The applicable stature and weight ranges for each size are shown in table XIII.

TABLE XIII. Stature and weight ranges.

Stature and Weight Range		
Size of Anti-G Garment	Stature Range (inches)	Weight Range (Pounds)
Small Regular	63 to 67-7/8	131 to 160
Small Long	68 to 72-7/8	131 to 160
Medium Regular	64-1/2 to 69-3/8	161 to 190
Medium Long	69-1/2 to 74-3/8	161 to 190
Large Regular	67 to 71-3/8	191 to 220
Large Long	71-1/2 to 75-3/8	191 to 220
Large Extra Long	75-1/2 to 79	191 to 230

6.5 Test samples. The test methods described in 4.6.3 for endurance and in 4.6.8 for bond strength (bladder), which are required for the sampling inspection specified in 4.5.3.2 and 4.5.3.3, are destructive tests. Since these tests are destructive, the Anti-G Garments and the bladders selected and tested in accordance with 4.5.3.2 (Plan A) and 4.5.3.3 (Plan B) should not be included in the quantities of serviceable finished end items required. The Anti-G Garments selected as samples and subjected to the endurance test should be delivered to the procuring

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activity. The bladders selected as samples and subjected to the bond strength (bladder) test should not be delivered.

6.6 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-81905 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Air Systems Command, Code 4.6.8.3, 48110 Shaw Road, Patuxent River, MD 20670.

6.7 Subject term (key word) listing.

Aircrew
Coverall
Protective clothing

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy - AS

Preparing activity:
Navy - AS

(Project 8475-2006-002)

NOTE: The activity listed above was interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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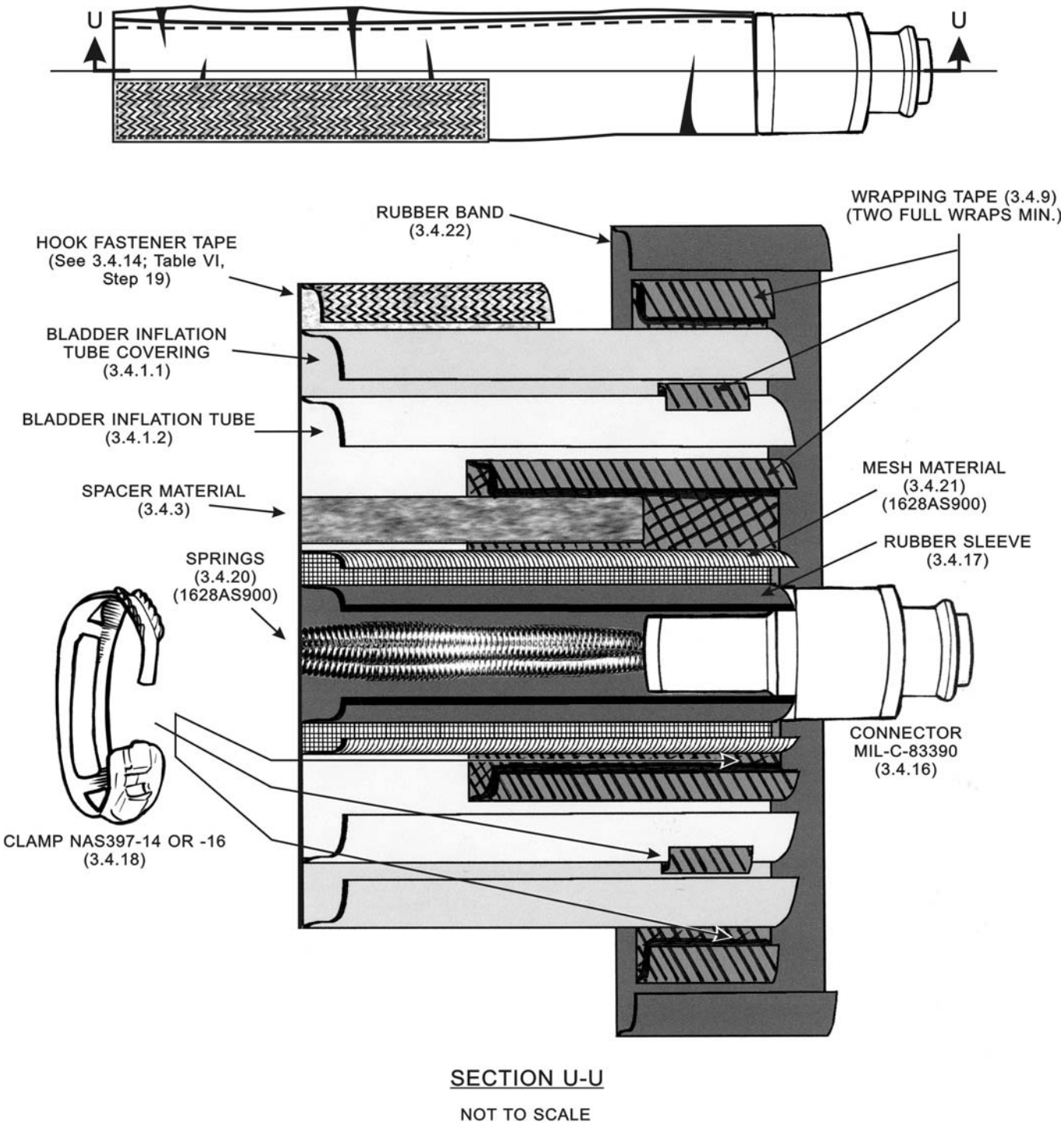


FIGURE 1. Inflation tube assembly.

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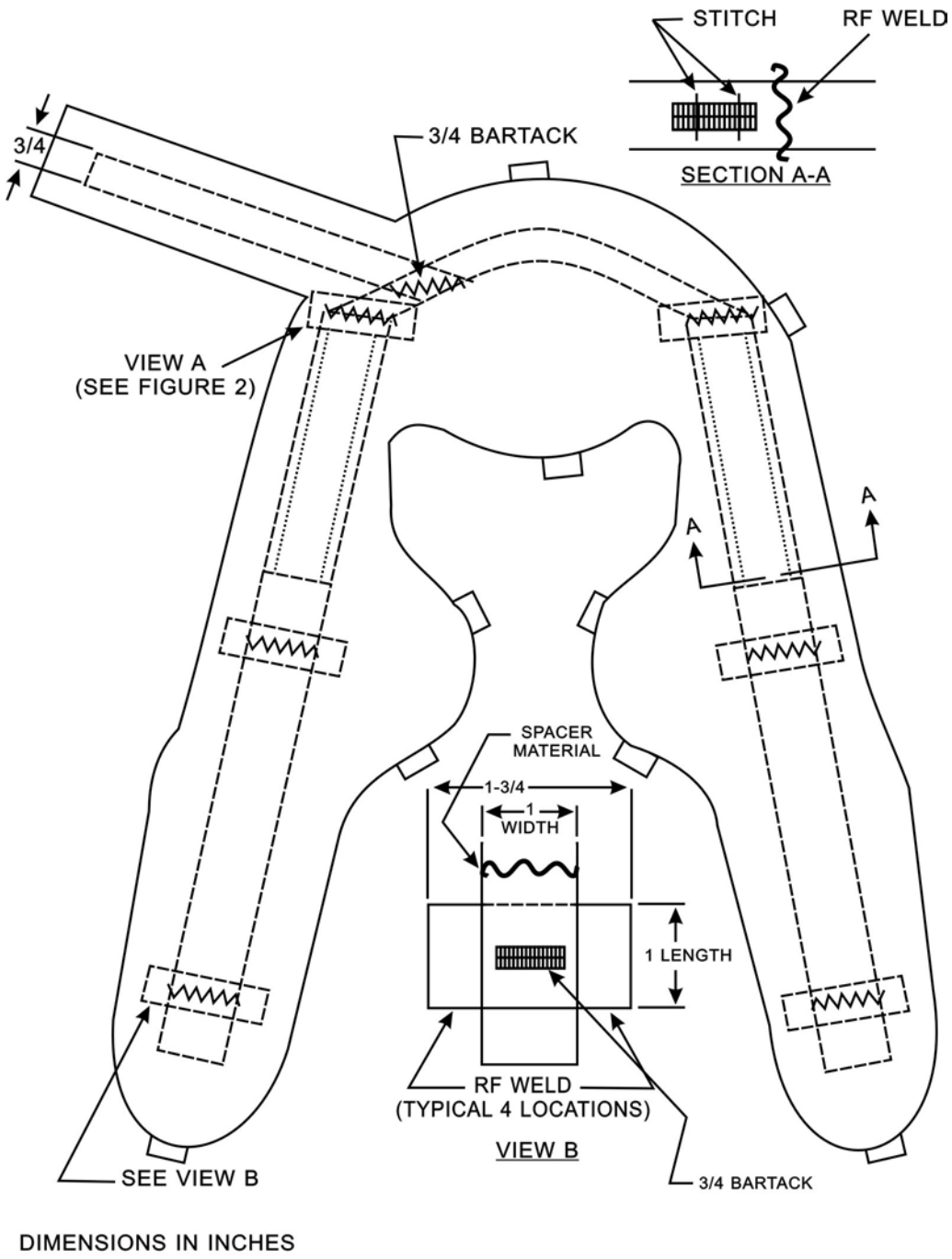


FIGURE 2. Bladder construction (units are inches).

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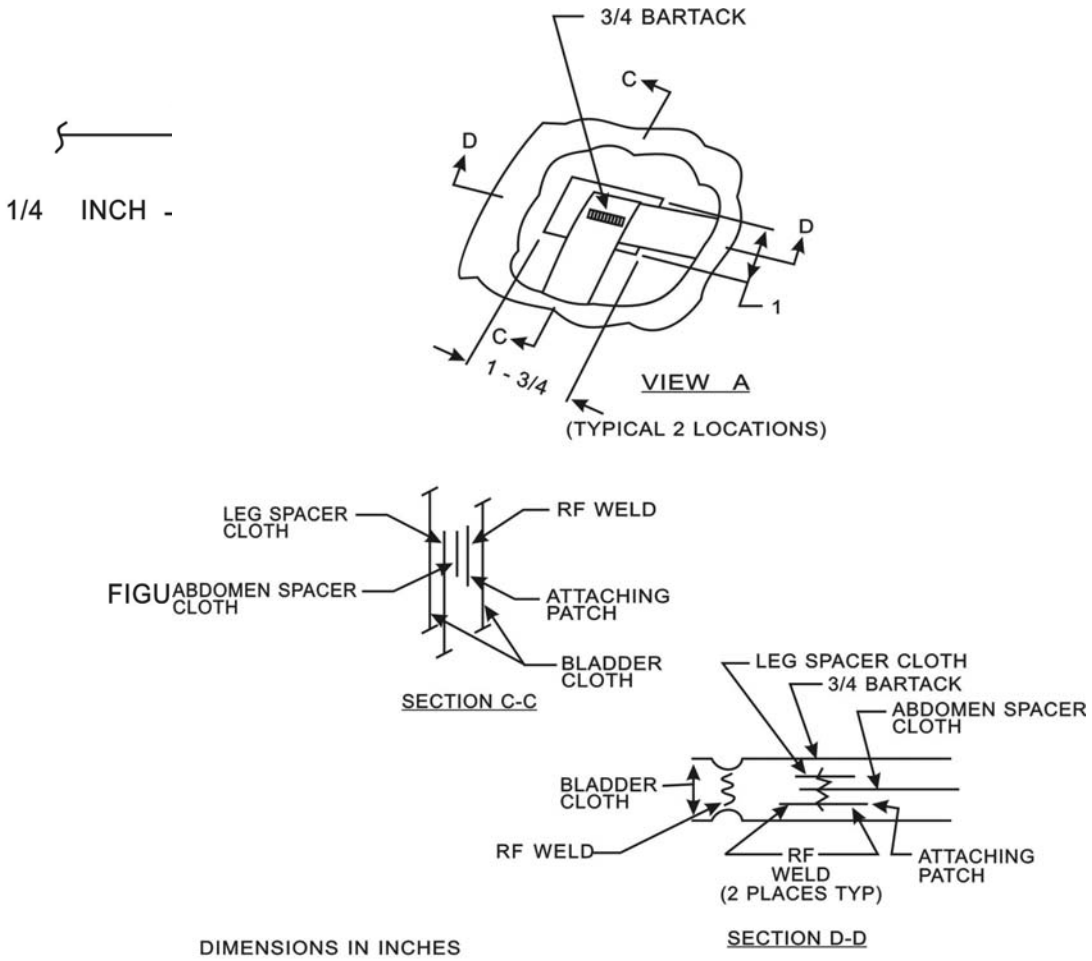


FIGURE 3. Bartack (View A).

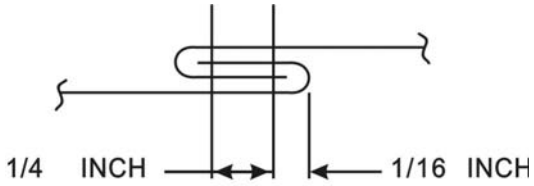


FIGURE 4. Seam type.

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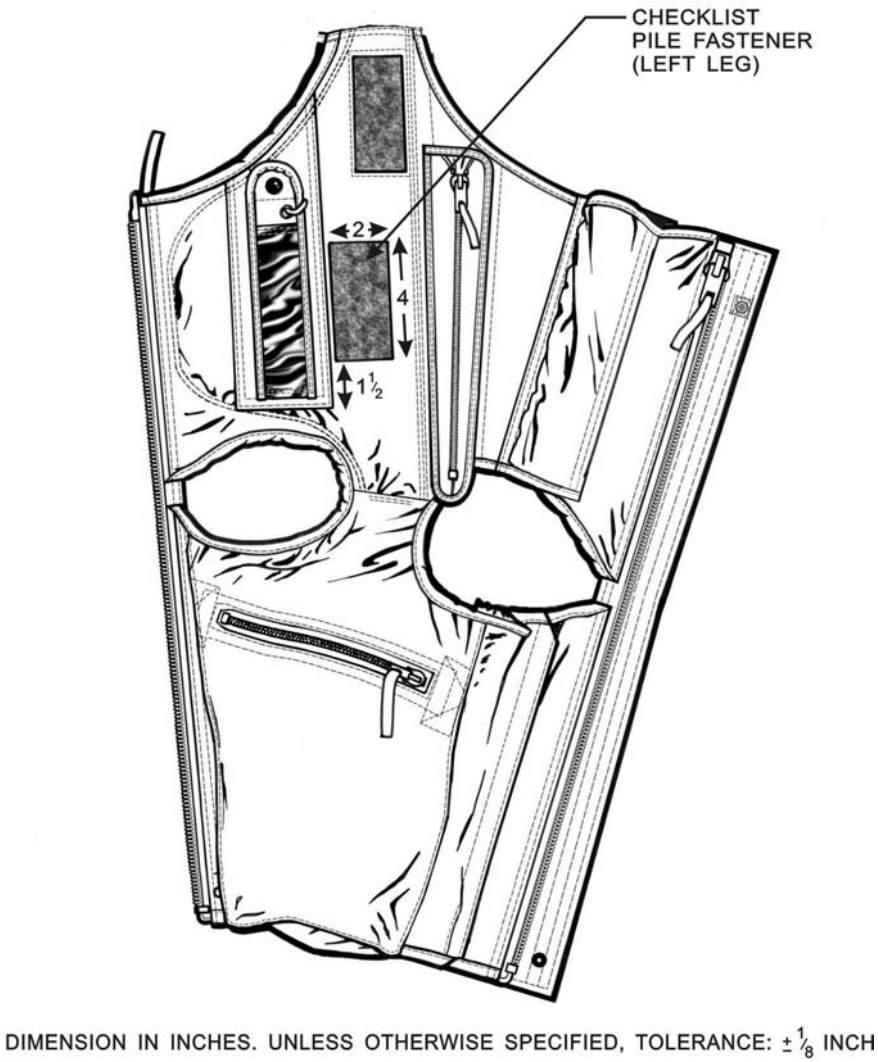


FIGURE 5. Location of checklist fastener.

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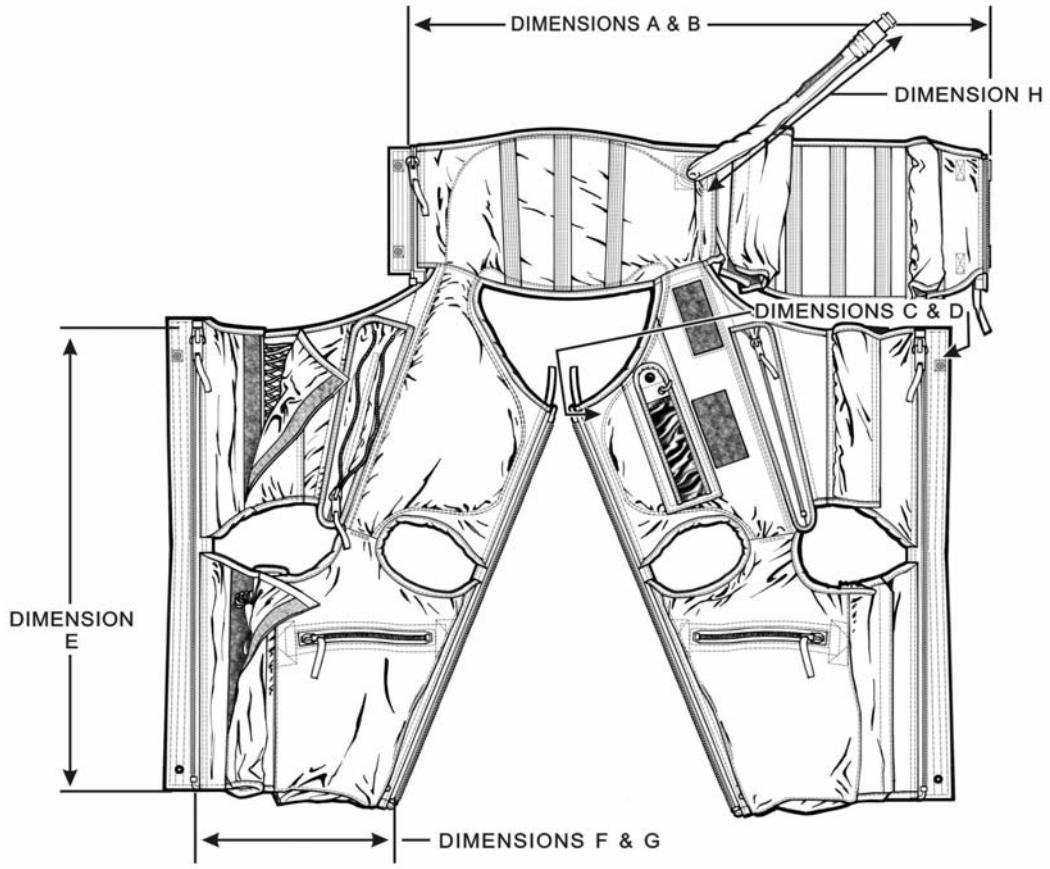


FIGURE 6. Finished measurements (see paragraph 4.7).

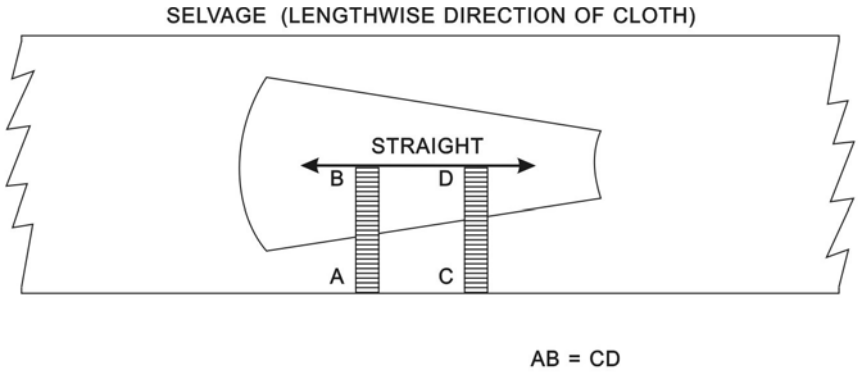


FIGURE 7. Directional line.

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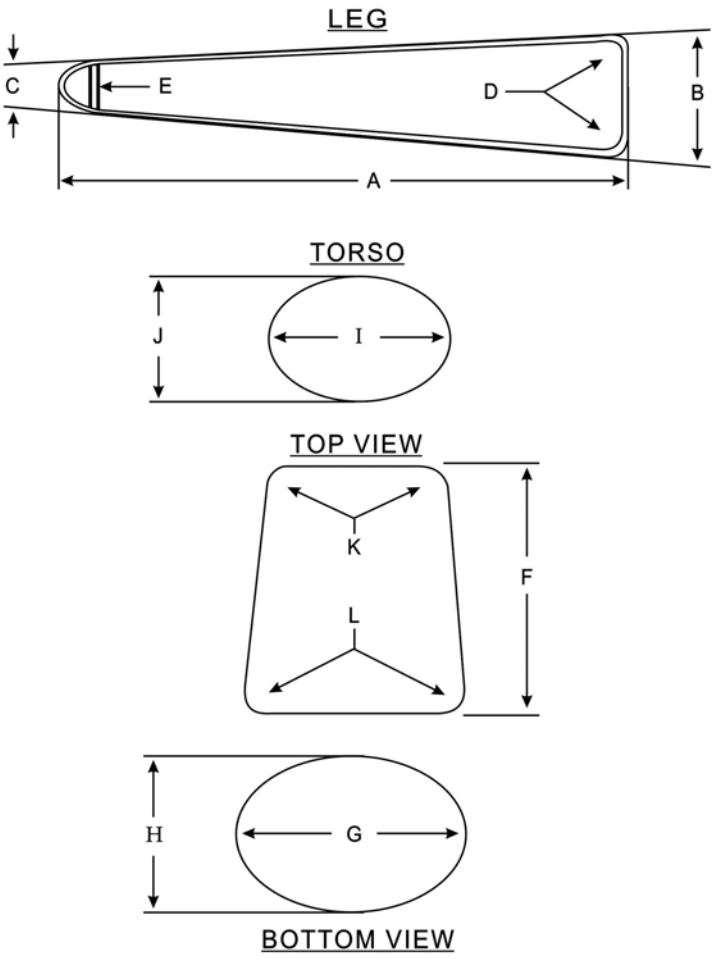


FIGURE 8. Models.

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